ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

ASSIGNMENT – 9

Apply Object Detection from CCTV Footage

Model: YOLOv8 Nano

1. Aim

To apply object detection to CCTV footage using a pretrained YOLOv8 model and identify objects in video frames using bounding boxes, class labels and confidence scores.

2. Objectives

- Understand object detection in video surveillance.
- Use a pretrained YOLOv8 Nano model.
- Detect multiple object classes in CCTV frames.
- Record confidence scores and bounding-box coordinates.
- Summarize detection counts and confidence.
- Generate an annotated output video.
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3. Video and Model Description

The supplied experiment uses a CCTV video named 12869951_3840_2160_30fps.mp4. The video has a resolution of 3840 × 2160 pixels, approximately 29.97 FPS, 528 frames and a duration of 17.62 seconds. YOLOv8 Nano (yolov8n.pt) is used with a confidence threshold of 0.35.

Property	Value
Model	YOLOv8 Nano
Resolution	3840 × 2160
FPS	29.97
Frames	528
Duration	17.62 s
Confidence threshold	0.35

4. Theory

Object detection is a computer-vision task that identifies objects in an image and simultaneously determines their locations. Unlike image classification, which assigns labels to an entire image, object detection produces a class label and a bounding box for each detected object.

YOLO, meaning You Only Look Once, is a family of one-stage object detectors designed for efficient detection. The model processes an image and predicts bounding boxes, object classes and confidence scores in a single inference pipeline.

YOLOv8 is a modern member of the YOLO family. The Nano version is designed to be relatively lightweight and suitable for fast inference compared with larger model variants.

A bounding box is commonly represented using coordinates such as x1, y1, x2 and y2, defining the top-left and bottom-right corners of the detected region. The coordinates allow the detected object to be localized within the frame.

The confidence score represents the model's confidence in a detection. A confidence threshold can be used to remove detections below a chosen score. A higher threshold usually produces fewer but more confident detections.

Non-Maximum Suppression is an important part of object detection pipelines. When multiple overlapping boxes represent the same object, redundant boxes can be suppressed so that the final result contains the most relevant detection.

Video object detection applies the image-detection process frame by frame. Detections can then be aggregated over time to analyze traffic, people, vehicles or other objects in surveillance footage.

CCTV object detection has applications in traffic monitoring, security surveillance, crowd analysis, parking systems, smart cities and automated incident monitoring. Performance can vary with camera angle, lighting, object size, motion blur and scene complexity.

The object detector used in this experiment is pretrained on a general-purpose object-recognition dataset. Consequently, the model can recognize common object categories without requiring the user to train a new detector from scratch.

Detection quality can be influenced by image resolution and computational resources. High-resolution CCTV footage contains more visual detail but also requires more computation. For practical surveillance systems, inference speed and detection accuracy must be balanced.

For video analysis, simply counting detections across all frames does not necessarily equal the number of unique physical objects because the same object may be detected repeatedly in many frames. The supplied detection count therefore represents detection instances, not unique tracked objects.

5. Methodology

- Load the CCTV video.
- Load the pretrained YOLOv8 Nano model.
- Read the video frame by frame.
- Run YOLO inference using a 0.35 confidence threshold.
- Record object class, confidence and bounding-box coordinates.
- Draw detections on each frame and save the annotated video.
- Aggregate detections by object class.
- Calculate average confidence and visualize detection statistics.

6. Outputs

6.1 Object Detection Summary

Object	Detection_Count	Average_Confidence
car	1683	0.5718
bus	75	0.4529
train	20	0.3947
truck	18	0.3964
person	12	0.4092
bench	5	0.5137
umbrella	2	0.3542

6.2 Representative CCTV Frame



Figure 1: Representative frame from the supplied CCTV output

6.3 Object Detection Counts

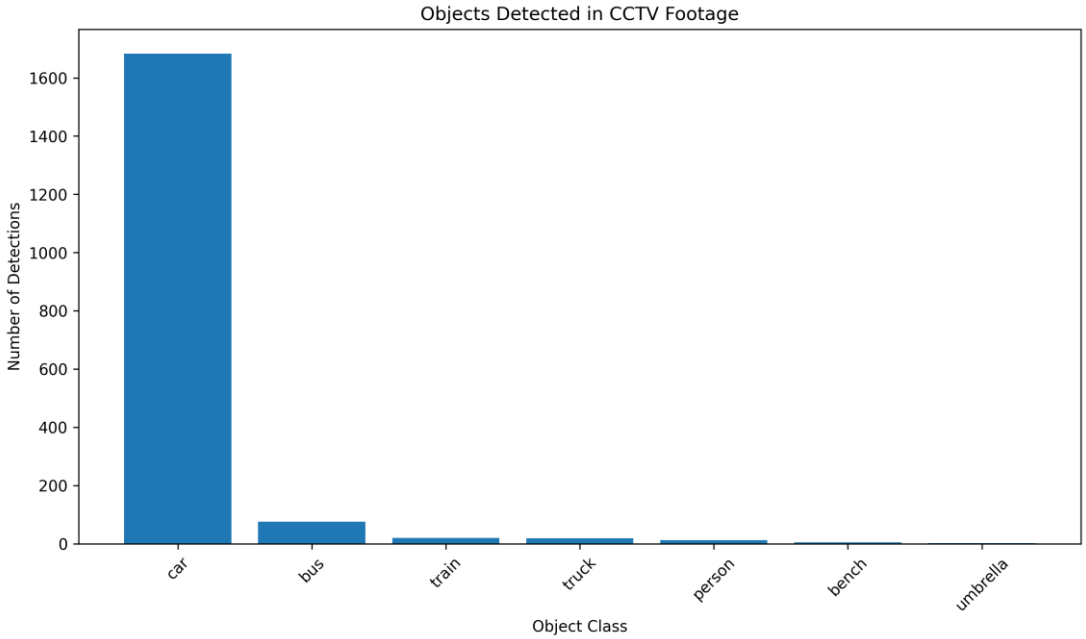


Figure 2: Number of detection instances by object class

6.4 Average Confidence

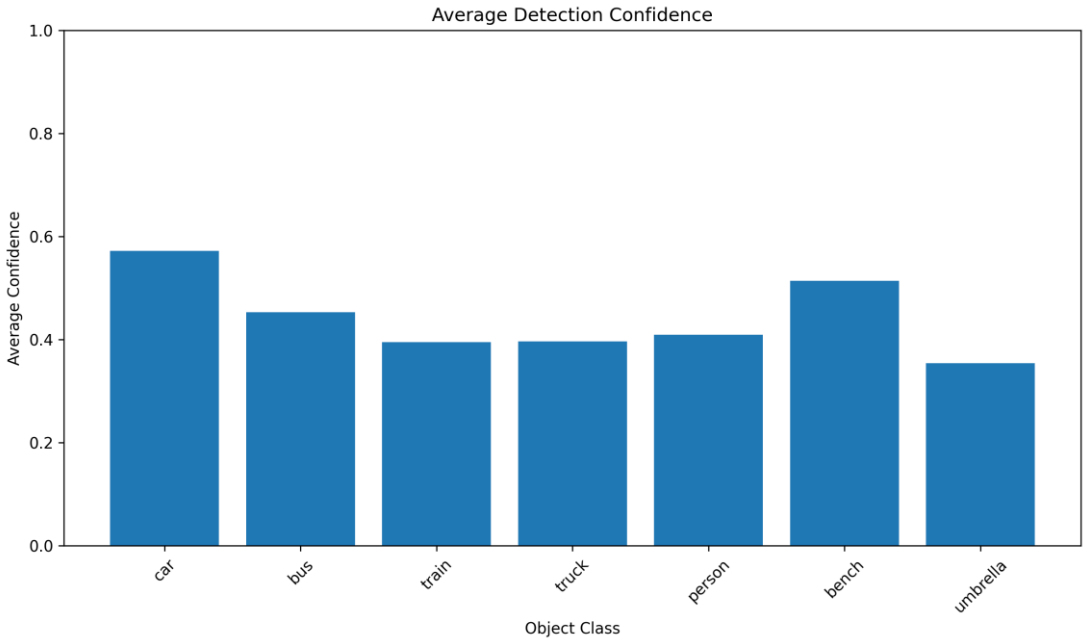


Figure 3: Average confidence by object class

6.5 Confidence Distribution

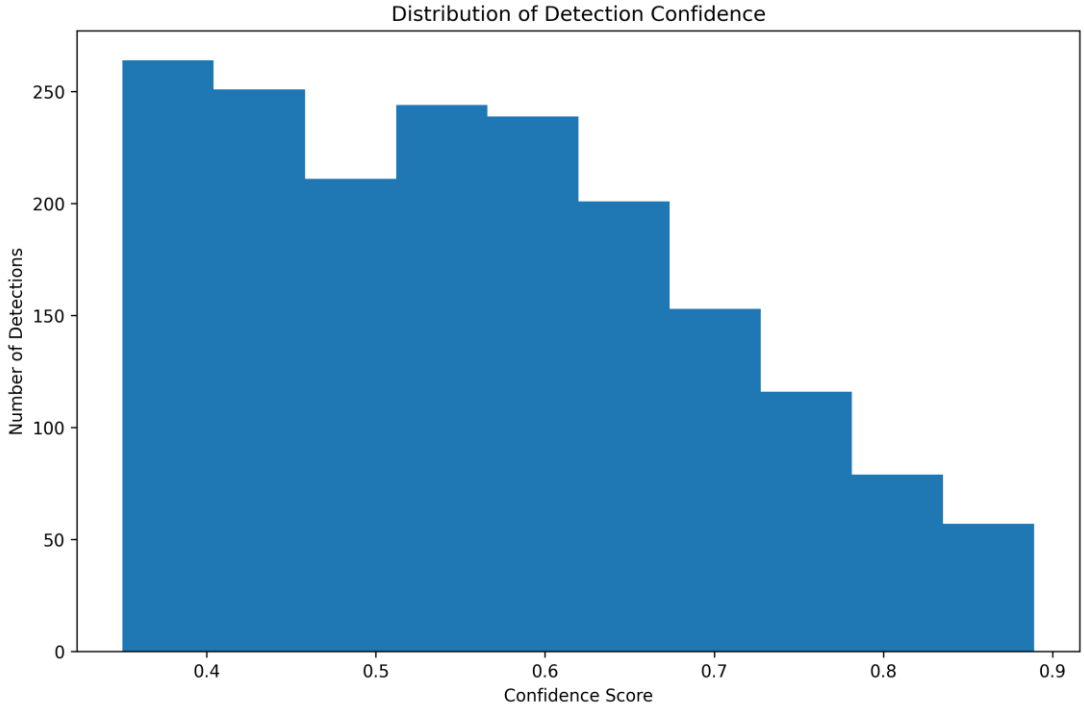


Figure 4: Distribution of detection confidence

6.6 Detections Across Frames

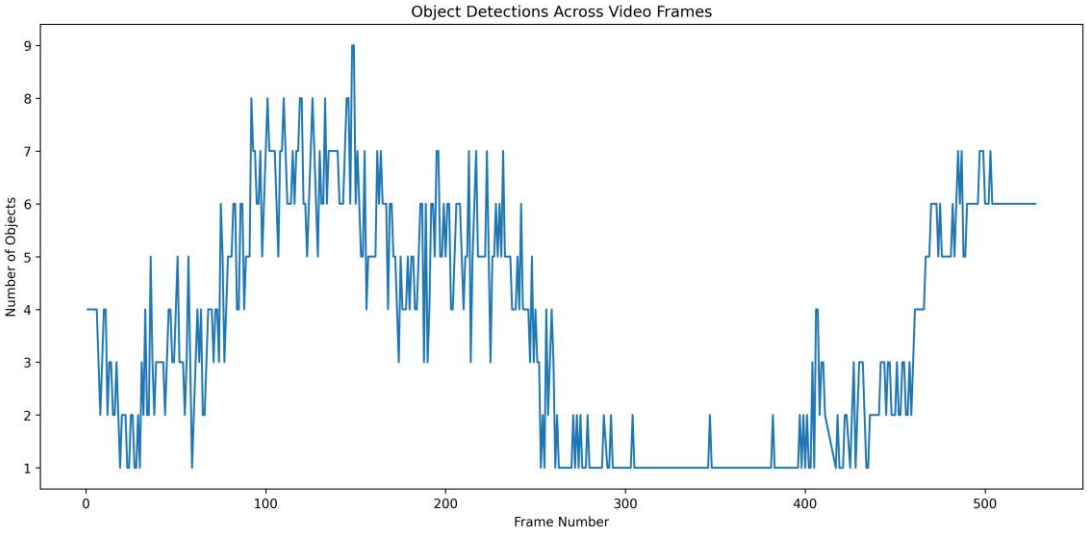


Figure 5: Detection instances across video frames

7. Result

YOLOv8 Nano successfully processed the supplied 17.62-second CCTV footage. A total of 1,815 detection instances were recorded across seven object classes. Cars were the dominant detected class with 1,683 detection instances. The overall average confidence was 0.5617.

8. Observations

- The model detected seven different object classes.
- Cars accounted for 1,683 of the 1,815 recorded detection instances.
- The confidence threshold was 0.35.
- The detected classes included car, bus, train, truck, person, bench and umbrella.
- Detection counts represent repeated frame-level detections and not unique tracked objects.
- CCTV detection performance can be affected by lighting, viewpoint, object size and video quality.

9. Conclusion

The experiment successfully demonstrated object detection from CCTV footage using YOLOv8 Nano. The system produced annotated video frames and quantitative detection statistics. The results show how pretrained object detectors can support automated surveillance and traffic-monitoring applications.